Check for updates

ACM DIGITAL LIBRARY

Association for Computing Machinery

acm open

Latest updates: https://dl.acm.org/doi/10.1145/3408318

PDF Download
3408318.pdf
21 March 2026
Total Citations: 44
Total Downloads: 1301

Published: 31 March 2021
Accepted: 01 June 2020
Revised: 01 May 2020
Received: 01 April 2020

Citation in BibTeX format

RESEARCH-ARTICLE

Gaussian Mixture Model Clustering with Incomplete Data

YI ZHANG, National University of Defense Technology China, Changsha, Hunan, China

MIAOMIAO LI, Changsha University, Changsha, Hunan, China

SIWEI WANG, National University of Defense Technology China, Changsha, Hunan, China

SISI DAI, National University of Defense Technology China, Changsha, Hunan, China

LEI LUO, National University of Defense Technology China, Changsha, Hunan, China

EN ZHU, National University of Defense Technology China, Changsha, Hunan, China

View all

Open Access Support provided by:

Zhejiang Normal University

National University of Defense Technology China

Chongqing University of Technology

Changsha University

ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), Volume 17, Issue 1s (January 2021)
https://doi.org/10.1145/3408318
EISSN: 1551-6865

Gaussian Mixture Model Clustering with Incomplete Data

YI ZHANG, School of Computer, NUDT
MIAOMIAO LI, Changsha University; School of Computer, NUDT
SIWEI WANG, SISI DAI, LEI LUO, and EN ZHU, School of Computer, NUDT
HUIYING XU, College of Mathematics and Computer Science, Zhejiang Normal University; Department of Computer Science, City University of Hong Kong
XINZHONG ZHU, College of Mathematics and Computer Science, Zhejiang Normal University
CHAOYUN YAO, State Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System, NUDT
HAORAN ZHOU, Chongqing University of Technology

Gaussian mixture model (GMM) clustering has been extensively studied due to its effectiveness and efficiency. Though demonstrating promising performance in various applications, it cannot effectively address the absent features among data, which is not uncommon in practical applications. In this article, different from existing approaches that first impute the absence and then perform GMM clustering tasks on the imputed data, we propose to integrate the imputation and GMM clustering into a unified learning procedure. Specifically, the missing data is filled by the result of GMM clustering, and the imputed data is then taken for GMM clustering. These two steps alternatively negotiate with each other to achieve optimum. By this way, the imputed data can best serve for GMM clustering. A two-step alternative algorithm with proved convergence is carefully designed to solve the resultant optimization problem. Extensive experiments have been conducted on eight UCI benchmark datasets, and the results have validated the effectiveness of the proposed algorithm.

CCS Concepts: • **Computing methodologies → Machine learning algorithms**; • **Theory of computation → Unsupervised learning and clustering**;

Additional Key Words and Phrases: GMM, clustering, EM, incomplete data

ACM Reference format:
Yi Zhang, Miaomiao Li, Siwei Wang, Sisi Dai, Lei Luo, En Zhu, Huiying Xu, Xinzhong Zhu, Chaoyun Yao, and Haoran Zhou. 2021. Gaussian Mixture Model Clustering with Incomplete Data. *ACM Trans. Multimedia Comput. Commun. Appl.* 17, 1s, Article 6 (March 2021), 14 pages.
https://doi.org/10.1145/3408318

This work was supported by the National Natural Science Foundation of China (project no. 61906020, 61701451, 61872377, and 61922088).
Authors' addresses: Y. Zhang, S. Wang, S. Dai, L. Luo, and E. Zhu, School of Computer, NUDT, China; M. Li (corresponding author), Changsha University; School of Computer, NUDT, China; email: miaomiaolinudt@gmail.com; H. Xu, College of Mathematics and Computer Science, Zhejiang Normal University; Department of Computer Science, City University of Hong Kong, China; X. Zhu, College of Mathematics and Computer Science, Zhejiang Normal University, China; C. Yao, State Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System, NUDT, China; H. Zhou, Chongqing University of Technology, China.
Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2021 Association for Computing Machinery.
1551-6857/2021/03-ART6 $15.00
https://doi.org/10.1145/3408318

1 INTRODUCTION

Clustering has been widely studied in the unsupervised learning community and used in various real-world applications. Plenty of excellent clustering algorithms have been proposed in the past few decades, such as K-Means, Fuzzy CMeans, DBSCAN, hierarchical clustering, and many more [13, 29]. These clustering algorithms show good clustering performance and are widely applied in simple data analysis situations; however, they cannot process complicated multi-modal data well, resulting in poor clustering performance [8, 22].

The Gaussian mixture model (GMM) clustering algorithm, which can achieve better clustering performance with complicated multi-modal data, is proposed to address this issue. GMM clustering has been extensively studied due to its effectiveness and efficiency [4] and widely applied in various fields, such as image segmentation problems in the fields of medicine and transportation [27].

Although the existing GMM clustering and its improved algorithms have achieved great success and demonstrated promising performance in various applications [19], these clustering algorithms all assume that the dataset is completely observable. However, in many practical applications, due to the absence of partial features among samples, this assumption may no longer hold, and thus the above algorithms cannot be effectively applied to cope with the absent features among data.

Many efforts have been devoted to addressing the problem of clustering incomplete data. Their main idea is to impute the incomplete features through filling algorithms first, such as zero-filling, mean-filling, k-nearest-neighbor filling, expectation-maximizing filling [10], and other improved algorithms [2, 7, 11, 15]. These methods try to reduce the negative impact of missing data by preprocessing and then apply standard GMM clustering algorithms to these imputed data matrices. However, the processes of data imputation and clustering are performed separately, resulting in the process of missing features' imputation cannot serve clustering tasks.

Therefore, different from existing approaches that first impute the absence and then perform GMM clustering tasks on the imputed data, we propose to integrate the imputation and GMM clustering into a unified learning procedure. Specifically, the missing data is filled by the result of GMM clustering, and the imputed data is then taken for GMM clustering. These two steps alternatively negotiate with each other to achieve optimum. By this way, the imputed data can best serve for GMM clustering. Further, a two-step alternative algorithm with proven convergence is carefully designed to solve the resultant optimization problem. Extensive experiments have been conducted on eight UCI benchmark datasets and the results have validated the effectiveness of the proposed algorithm.

The contributions of this article are summarized as follows:

- Different from the existing algorithms that use imputing as preprocessing and clustering, respectively, we unify these two processes into one optimization goal. Missing features are dynamically imputed to better serve the clustering tasks, while existing observed entries are guaranteed to remain unchanged.
- According to our knowledge, this is the newest work proposed to extend the Gaussian mixture model to cope with a clustering task with incomplete data. We represent the formulation and fulfill the aforementioned ideas. In addition, we designed an alternating optimization algorithm to solve the optimization problem of clustering with incomplete data.
- Extensive experimental research is conducted on eight UCI benchmark datasets. As noted, our algorithm is consistent to achieve state-of-the-art performance, compared to other filling methods. Experiment results have validated the effectiveness of the proposed algorithm.

2 RELATED WORK

2.1 Gaussian Mixture Model

Given a data matrix $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_m\}$, the maximum likelihood estimation method can be applied to solve its model parameters [4], and the objective formulation is represented as follows:

$$\max_{\boldsymbol{\alpha}, \boldsymbol{\mu}, \boldsymbol{\Sigma}} \sum_{j=1}^{n} \ln\left(\sum_{i=1}^{k} \alpha_i \frac{1}{2\pi^{n/2}|\boldsymbol{\Sigma}_i|^{1/2}} e^{-\frac{1}{2}(\boldsymbol{x}_j-\boldsymbol{\mu}_i)^T\boldsymbol{\Sigma}_i^{-1}(\boldsymbol{x}_j-\boldsymbol{\mu}_i)}\right), \tag{1}$$

where $\boldsymbol{\mu}_i$ and $\boldsymbol{\Sigma}_i$ are the parameters of the ith Gaussian mixture component, and α_i is the corresponding mixture coefficient, which satisfies $\sum_{i=1}^{k} \alpha_i = 1$.

Although the existing GMM clustering and its improved algorithms have achieved great success and demonstrated promising performance in various applications, these clustering algorithms all assume that the dataset is completely observable. However, in many practical applications, due to the absence of partial features among samples, the above algorithms cannot be effectively applied to cope with the absent features among data.

Many efforts have been devoted to addressing the problem of clustering incomplete data. These methods try to reduce the negative impact of missing data by pre-processing, and then apply standard GMM clustering algorithms to these imputed data matrices.

2.2 Imputation Methods

2.2.1 Basic Statistical Methods. The statistical methods seek more useful information from the missing data. Most of them impute the absent values through statistical properties, rather than discarding incomplete information. They fill the incomplete entries with constants to take a complete data sample and apply it to learning tasks.

For example, the simplest and most common methods used filling values like zero, conditional mean, and median in this dimension. Specially, the KNN-filling method has been proposed to impute the missing entries with the mean feature on the K-closest neighbors in that dimension [9]. But it does not work when the missing data ratio is relatively large, and thus we do not use it as a comparison algorithm.

Different from the aforementioned methods, another statistical class to deal with incomplete features is followed by Bayesian frameworks. These frameworks are often shown in a maximum-likelihood manner, which imputes the missing values with the most likely estimated numbers. The most followed or popular method is the expectation maximization (EM) algorithm [5].

2.2.2 Dynamically K-means Filling. Given data matrix $\boldsymbol{X} = \{\boldsymbol{x}_i\}_{i=1}^{n}$ and the number of clusters k, we have three variables to be optimized: the data matrix $\boldsymbol{X}$, assignment matrix $\boldsymbol{H}$, and the clusters' centers $\boldsymbol{\mu}_c (1 \le c \le k)$[23]. By imposing the constraint on $\boldsymbol{X}$, the formulation can be expressed as follows:

$$\begin{aligned} &\min_{\boldsymbol{H}, \{\boldsymbol{\mu}_c\}_{c=1}^{k}, \boldsymbol{X}} \sum_{i=1}^{k}\sum_{c=1}^{k} \boldsymbol{H}_{ic}|\boldsymbol{x}_i - \boldsymbol{\mu}_c|^2 \\ &s.t. \quad \forall i, \sum_{c=1}^{k} \boldsymbol{H}_{ic} = 1, \ \boldsymbol{x}_i(o_i) = \boldsymbol{x}_i^o, \end{aligned} \tag{2}$$

where $n_c = \sum_{i=1}^{n} \boldsymbol{H}_{ic}$ and $\boldsymbol{\mu}_c = \frac{1}{n_c}\sum_{i=1}^{n} \boldsymbol{H}_{ic}\boldsymbol{x}_i$ are the sample number and the centroid of the cth $(1 \le c \le k)$ cluster and $\boldsymbol{x}_i(o_i)$ represents the observable elements of the ith sample $\boldsymbol{x}_i$. Moreover, we also impose constraints on the observable part of data matrix $\boldsymbol{x}_i(o_i)$ to ensure their values are kept unchanged during the optimization process.

2.3 Other Clustering Algorithms

2.3.1 Multi-view Clustering. Multi-view clustering (MVC) integrates the available multi-view information to categorize data items with similar structures or patterns into the same group, aiming to optimally integrate features from different views to improve clustering performance [3]. It has been intensively studied and widely applied into various applications during the last few decades [6, 14, 16, 30].

The work in [24] presented a novel approach towards subspace clustering over multi-view data, which achieves the data correlation consensus on multi-views through an angular-based regularizer. The work in [25] proposes to learn a clustered low-rank representation via structured matrix factorization for multi-view spectral clustering. The work in [16] proposed a multiple kernel k-means clustering algorithm with matrix-induced regularization to reduce the redundancy of the pre-defined kernels. A multiple kernel algorithm is proposed to allow the optimal kernel to reside in the neighborhood of the combinational kernels [31]. The works in [17, 18] proposed simple yet effective algorithms to address the incomplete kernel matrices. These newly proposed methods greatly improve the practical application ability of multi-view clustering; the study of multi-view clustering will be a direction worthy of continued exploration and research.

2.3.2 Deep Clustering. Deep learning can automatically extract abstract, non-linear, and cluster-friendly features from complex data structures, thereby improving the performance of the algorithm. In recent years, it has made great efforts in various application fields. Clustering algorithms are supplemented by deep neural network feature learning in order to capture the data itself or its internal structure to better separate clusters [1, 20, 21].

The work in [26] further proposed the concept of cyclic consistency loss on the basis of adversarial training in order to strengthen the correlation between the input and the corresponding output. It provides a good inspiration for the research on deep neural network clustering.

The work in [12, 28] studied and improved deep embedding clustering (DEC). DEC is one of the most representative deep clustering methods, which has gained a lot of attention in the field of deep clustering. Using the autoencoder as the network architecture, clustering is performed by minimizing the reconstruction loss of the autoencoder and the KL divergence between the label distribution and the auxiliary distribution. It has achieved encouraging results and has become one of the references for the performance of newer deep clustering algorithms.

3 GMM CLUSTERING WITH INCOMPLETE DATA

The processes of data imputation and clustering are performed separately in the above methods, resulting in the process of missing features' imputation cannot serve clustering tasks. Therefore, different from existing approaches that first impute the absence and then perform GMM clustering tasks on the imputed data, we propose to integrate the imputation and GMM clustering into a unified learning procedure. Specifically, the missing data is filled by the result of GMM clustering, and the imputed data is then taken for GMM clustering. These two steps alternatively negotiate with each other to achieve optimum. By this way, the imputed data can best serve for GMM clustering.

For incomplete data, each sample $\boldsymbol{x}_j (1 \leq j \leq n)$ can be divided into two parts: the observable features $x_j(o_j)$ and missing features $\boldsymbol{x}_j(m_j)$. When we optimize the prior of k components, the mean $\boldsymbol{\mu}_k$, and the covariance $\boldsymbol{\Sigma}_k$ for the kth Gaussian basis, we propose to optimize the missing parts of the data $\boldsymbol{x}_j(m_j)$ while keeping the observable features $\boldsymbol{x}_j(o_j)$ unchanged during the optimization process.

3.1 The Proposed Formulation

Based on the aforementioned discussion, we redefine the objective of the Gaussian mixture model to cope with incomplete data clustering. Given data matrix $\boldsymbol{X}$ and the number of clusters k, we have four variables to be optimized: the data matrix $\boldsymbol{X}$, the prior of k components, the mean $\boldsymbol{\mu}_k$, and the covariance $\boldsymbol{\Sigma}_k$. By imposing the constraint on the observable features of the data, we need to optimize the model parameters $\{\boldsymbol{\alpha}, \boldsymbol{\mu}, \boldsymbol{\Sigma}\}$ and the missing data $\boldsymbol{x}_j(m_j)$. Thus, we set our Gaussian mixture model clustering with incomplete data as follows:

$$\max_{\boldsymbol{\alpha}, \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{X}} \sum_{j=1}^{n} \ln\left(\sum_{i=1}^{k} \alpha_i p(\boldsymbol{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)\right) \quad s.t.\ \boldsymbol{x}_j(o_j) = \boldsymbol{x}_j^o, \tag{3}$$

where $\boldsymbol{\Sigma}_i$ is a symmetric positive definite matrix, $\boldsymbol{x}_j(o_j)$ represents the observable elements of the jth sample $\boldsymbol{x}_j$, and $p(\boldsymbol{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ represents the probability density of the jth sample value corresponding to the ith Gaussian component, which can be expressed as follows:

$$p(\boldsymbol{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) = \frac{1}{2\pi^{n/2}|\boldsymbol{\Sigma}_i|^{1/2}} e^{-\frac{1}{2}(\boldsymbol{x}_j - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1} (\boldsymbol{x}_j - \boldsymbol{\mu}_i)}. \tag{4}$$

Moreover, we also impose constraints on the observable part of data matrix $\boldsymbol{x}_j(o_j)$ to ensure their values are kept unchanged during the optimization process. And we can continuously optimize the missing part of the data matrix $\boldsymbol{x}_j(m_j)$ through the EM algorithm during the clustering process.

3.2 Alternative Optimization

As seen, the additional constraint $\boldsymbol{x}_i(o_i) = \boldsymbol{x}_i^o$ makes the whole optimization problem difficult to solve. In order to solve it, we design a two-step alternative optimization algorithm with a fast convergence rate, where the first step can be easily solved by applying the existing off-the-shelf packages and the second step can be solved by the matrix derivative method. To solve the model's parameters $\{(\alpha_i, \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i | 1 \le i \le k)\}$, we can use maximum likelihood estimation, which maximizes the log-likelihood as follows:

$$LL(\boldsymbol{X}) = \sum_{j=1}^{n} \ln\left(\sum_{i=1}^{k} \alpha_i p(\boldsymbol{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)\right). \tag{5}$$

And we use the EM algorithm for iterative optimization.

3.2.1 Expectation Step. For the dataset $X = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_m\}$, let the random variable $z_j \in \{1, 2, \ldots, k\}$ denote the Gaussian mixture component to which the sample x_j belongs, and its value is unknown. Obviously, the prior probability of z_j corresponds to α. According to Bayes' theorem, the posterior distribution of z_j corresponds to

$$\gamma_{ji} = \frac{\alpha_i p(\boldsymbol{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}{\sum_{l=1}^{k} \alpha_l p(\boldsymbol{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_l)}. \tag{6}$$

In each iteration, first calculate the posterior probability γ_{ji}; the probability for each sample belongs to each Gaussian mixture component according to the current parameters, which is the Expectation Step (E-Step).

3.2.2 Maximization Step.

ALGORITHM 1: Gaussian Mixture Model Clustering with Incomplete Data

Input: incomplete data matrix $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_m\}$, number of cluster k, convergence tolerance ϵ, and max iteration times $maxIter$
Parameter: mixed coefficients $\boldsymbol{\alpha}$, mean vector $\boldsymbol{\mu}$, covariance matrix $\boldsymbol{\Sigma}$, and data matrix $\boldsymbol{X}$
Output: complete data matrix $\boldsymbol{X}$ and Cluster division C

```
Initialize the missing values through basic imputing method.
Initialize the parameters of Gaussian Mixture Model according to the data matrix X.
repeat
    Calculate the posterior probability Γ by solving Equation (6) that each sample generated by each Gaussian mixed component.
    Update μ_i, Σ_i, α_i by solving Equations (9)–(11) with fixed other GMM parameters and data matrix X. (1 ≤ i ≤ k), respectively.
    Update X by solving Equation (14) with fixed mixed coefficients α, mean vector μ, and covariance matrix Σ.
    Iteration times add one : t = t + 1.
until (obj^(t) − obj^(t−1))/obj^(t) ≤ ε or t ≥ maxIter
C_i = ∅ (1 ≤ i ≤ k).
for j = 1, 2, ..., n do
    Determine cluster label for sample x_j :
    λ_j = arg max_{i∈{1,2,...,k}} γ_ji.
    Divide x_j into corresponding clusters:
    C_{λ_j} = C_{λ_j} ∪ {x_j}.
end for
```

3.2.3 Optimizing Parameters α_i, $\boldsymbol{\mu}_i$, and $\boldsymbol{\Sigma}_i$ with fixed $\boldsymbol{X}$. At first, the optimization Equation (3) can be divided into k sub-problems with consideration to each component of the Gaussian mixture model:

$$\max_{\boldsymbol{\alpha}, \boldsymbol{\mu}, \boldsymbol{\Sigma}} \sum_{j=1}^{n} \ln\left(\alpha_i p(\boldsymbol{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)\right) \quad s.t.\ \boldsymbol{x}_j(o_j) = \boldsymbol{x}_j^o. \tag{7}$$

Optimizing $\boldsymbol{\mu}_i$ with fixed α_i, $\boldsymbol{\Sigma}_i$, and $\boldsymbol{X}$:

With mixed coefficients α_i and covariance matrices $\boldsymbol{\Sigma}_i$ being fixed, we make the partial derivative of Equation (7) for $\boldsymbol{\mu}_i$ equal to zero, and obtain:

$$\sum_{j=1}^{n} \frac{\alpha_i p(\boldsymbol{x}_j | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}{\sum_{l=1}^{k} \alpha_i p(\boldsymbol{x}_j | \boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)} (\boldsymbol{x}_j - \boldsymbol{\mu}_i) = 0, \tag{8}$$

$$\boldsymbol{\mu}_i = \frac{\sum_{j=1}^{n} \gamma_{ji} \boldsymbol{x}_j}{\sum_{j=1}^{n} \gamma_{ji}}. \tag{9}$$

Optimizing $\boldsymbol{\Sigma}_i$ with fixed α_i, $\boldsymbol{\mu}_i$, and $\boldsymbol{X}$: Similar to the previous step, we make the partial derivative of Equation (7) for $\boldsymbol{\Sigma}_i$ equal to zero, and obtain:

$$\boldsymbol{\Sigma}_i = \frac{\sum_{j=1}^{n} \gamma_{ji} (\boldsymbol{x}_j - \boldsymbol{\mu}_i)(\boldsymbol{x}_j - \boldsymbol{\mu}_i)^T}{\sum_{j=1}^{n} \gamma_{ji}}. \tag{10}$$

Optimizing α_i with fixed $\boldsymbol{\mu}_i$, $\boldsymbol{\Sigma}_i$, and $\boldsymbol{X}$: For mixed coefficients α_i, because of $\alpha_i \geq 0$, $\sum_{i=1}^{k} \alpha_i = 1$, consider the Lagrangian form, $LL(X) + \lambda(\sum_{i=1}^{k} \alpha_i - 1)$, where λ is a Lagrangian multiplier, and

obtain the new value of α_i by making the derivative of the formula for α_i be zero:

$$\alpha_i = \frac{1}{m}\sum_{j=1}^{n}\gamma_{ji}. \tag{11}$$

3.2.4 Optimizing Data Matrix $\boldsymbol{X}$ with Fixed $\boldsymbol{\alpha}$, $\boldsymbol{\mu}$, and $\boldsymbol{\Sigma}$. The problem in Equation (3) can be divided into n individual sub-problems with consideration to each sample $\boldsymbol{x}_j$; the optimization can be equivalently rewritten as follows:

$$\max_{\boldsymbol{X}} \sum_{i=1}^{k} \alpha_i p(\boldsymbol{x}_j|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \quad s.t.\ \boldsymbol{x}_j(o_j) = \boldsymbol{x}_j^o, \tag{12}$$

where $\boldsymbol{x}_j(o_j)$ represents the observable elements of the jth sample $\boldsymbol{x}_j$. Constraints are imposed on the observable part of data matrix $\boldsymbol{x}_j(o_j)$ to ensure their values are kept unchanged during the optimization process.

To solve the difficult problem, calculating the partial derivative of Equation (12) for $\boldsymbol{x}_m$, we divide the mean $\boldsymbol{\mu}_i$ and the covariance matrices $\boldsymbol{\Sigma}_i$ the same as the data matrices into two parts, the observable and missing parts, as follows:

$$\boldsymbol{x}_j = \begin{bmatrix}\boldsymbol{x}_{jo} & \boldsymbol{x}_{jm}\end{bmatrix},\ \boldsymbol{\mu}_i = \begin{bmatrix}\boldsymbol{\mu}_{io} & \boldsymbol{\mu}_{im}\end{bmatrix},\ \boldsymbol{\Sigma}_i^{-1} = \begin{bmatrix}\boldsymbol{\Sigma}_{ioo}^{-1} & \boldsymbol{\Sigma}_{iom}^{-1} \\ \boldsymbol{\Sigma}_{imo}^{-1} & \boldsymbol{\Sigma}_{imm}^{-1}\end{bmatrix}. \tag{13}$$

We make the partial derivative of Equation (12) for $\boldsymbol{x}_m$ to zero, and finally we can get an analytic solution as follows:

$$\boldsymbol{X}_m = \left(\sum_{i=1}^{k} \alpha_i P_i \boldsymbol{\Sigma}_{imm}^{-1}\right)^{-1} \sum_{i=1}^{k} \alpha_i P_i \left(\boldsymbol{\Sigma}_{imo}^{-1}\boldsymbol{\mu}_{io} + \boldsymbol{\Sigma}_{imm}^{-1}\boldsymbol{\mu}_{im} - \boldsymbol{\Sigma}_{imo}^{-1}\boldsymbol{x}_{jo}\right), \tag{14}$$

where $P_i = p(\boldsymbol{x}_j|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$. As seen from Equation (14), the missing elements of each sample $\boldsymbol{x}_j$ are imputed with the corresponding dimension of the Gaussian mixture component and the observable features of the sample.

According to the posterior probability γ_{ji} calculated in the E-Step, the model parameters and missing data values are updated by maximizing the likelihood estimation method, which is the Maximization Step (M-Step).

3.3 Discussion and Extension

Computational complexity: Compared to the basic GMM clustering algorithm, our Algorithm 1 considers the data matrix X as another variable to be optimized. In Equation (14), we update the missing values by maximizing the likelihood estimation objects with the GMM parameters, and when updating the missing values, we need to invert the covariance matrix whose time complexity is o(d^3). Therefore, the time complexity of our algorithm is O($tknd^3$), where t, k, n, d represent the number of iterations, clusters, samples, and dimensions, respectively.

Extensions: The algorithm in this work can be extended from the following aspects. First, the initialization parameters of GMM can be readily extended to be obtained through other methods for serving clustering better. For example, we can consider the denseness of the initial data distribution and the impact of outliers. And the idea of joint imputation and clustering is so natural that it can be generalized to other learning tasks such as classification, feature selection/extraction, etc. The one-stage framework can also be extended to handle noise by dynamically clustering.

Table 1. Datasets Used in Our Experiments

Dataset	#Samples	#Dimensions	#Clusters
Iris	150	4	3
AlcoholQCM	125	10	5
Seeds	210	7	3
Wine	178	13	3
Segment	2,310	18	7
ElectricalGrid	10,000	13	2
Avila	20,871	10	12
Letter	20,000	16	26

4 EXPERIMENT

4.1 Experiment Setting

The proposed algorithm is experimentally evaluated on eight widely used UCI datasets, including Iris, AlcoholQCM, Seeds, Wine, Segment, ElectricalGrid, Avila, and Letter. Table 1 shows the details of the datasets.

The incompleteness is randomly generated by the original complete data, which can be publicly downloaded from the UCI Machine Learning Repository. The number of samples varies from 100 to over 20,000, and dimensions from 4 to 18.

Iris: It is perhaps the best-known database to be found in the pattern recognition literature. The dataset contains three types of iris plant with a total of 150 instances. The sample features are four real numeric attributes of sepal length, sepal width, petal length, and petal width.

AlcoholQCM: For different types of alcohol, QCM gas sensors with two channels are used to measure gas with five different air-gas mixture ratios. There are 125 samples, and each sample has 10 real numeric attributes. There are five types of alcohol: 1-octanol, 1-propanol, 2-butanol, 2-propanol, and 1-isobutanol.

Seeds: It is derived from the grains of three different wheat varieties combined harvested in experimental fields. A soft X-ray technique and GRAINS package were used to construct all seven, real-valued attributes. There were a total of 210 samples.

Wine: These data are the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. The analysis determined the quantities of 13 constituents found in each of the three types of wines. There are a total of 178 samples.

Segment: The instances were drawn randomly from a database of outdoor images. The images were hand-segmented to create a classification for every pixel. Each sample is a 3x3 region, and there are a total of 2,310 samples with 19 real numeric attributes.

ElectricGrid: It is the data of the local stability analysis of the four-node star system implementing the decentral smart grid control concept. It has a total of 10,000 samples, which are divided into two clusters. The sample features are 13 real numeric attributes.

Avila: It has been extracted from 800 images of the "Avila Bible," a twelfth-century giant Latin copy of the Bible. There are a total of 20,871 samples, which are divided into 12 clusters. The sample features are 10 real numeric attributes such as spacing and margin.

Letter: It consists of hand-written digits with a total of 20,000 samples. The character images were based on 20 different fonts and each letter within these 20 fonts was randomly distorted to produce a file of 20,000 unique stimuli. Each stimulus was converted into 16 primitive numerical attributes, which were then scaled to fit into a range of integer values from 0 through 15. The labels are 26 capital letters in the English alphabet.

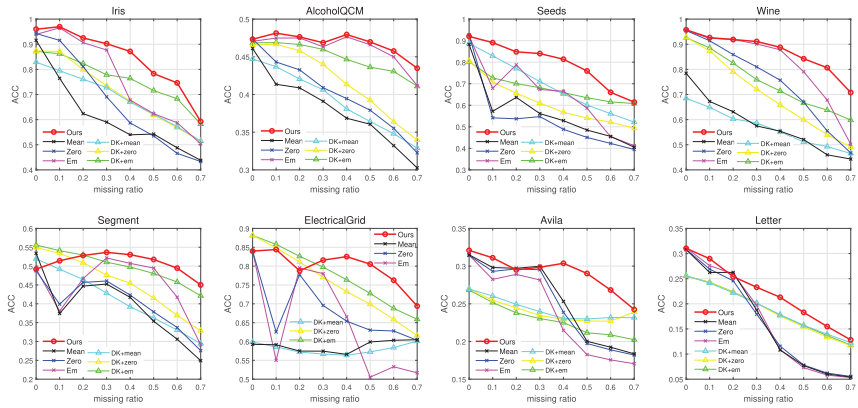

Fig. 1. ACC comparison with the variation of missing ratios on eight benchmark datasets.

We compare the proposed Dynamically GMM Clustering algorithm with several commonly used imputation methods, including mean filling (MF), zero filling (ZF), EM filling. In addition, we compare with the recently proposed dynamically K-means filling (DK) combined with the previous three methods.

For all the datasets, it is assumed that the true number of clusters k is known and it is set as the number of classes. We randomly generate the incompleteness by the original complete data matrix and the missing ratios of generated matrices are 10%, 20%, 30%, 40%, 50%, 60% and 70%, which affects the performance of the algorithms to varying degrees. To show this point in depth, we compare these algorithms with respect to the missing ratio.

The widely used clustering accuracy (ACC), normalized mutual information (NMI), F-score, and purity (PUR) are applied to evaluate the clustering performance of each algorithm.

For all algorithms, we repeat each experiment 50 times with random initialization to reduce the effect of the randomness of the initial value selection of the K-means and GMM clustering algorithm and report the average result. Meanwhile, we randomly generated the incompleteness patterns 20 times in the above-mentioned way and report the statistical results.

The aggregated ACC, NMI, F-score, and PUR are used to evaluate the performance of the algorithms in comparison. Taking the aggregated ACC, for example, it is obtained by averaging the averaged ACC achieved by an algorithm over different missing ratios. Our implementation is based on MATLAB and is available at https://github.com/Zhangyi1231/GMM-with-Incomplete-Data.

4.2 Clustering Performance

Figures 1 , 2, 3, and 4 present the ACC, NMI, PUR, and F-score, respectively, in comparison with the above algorithms with different missing ratios on all datasets. We have the following observations:

(1) The proposed algorithm significantly and consistently outperforms existing two-stage imputation methods. For example, it exceeds the best two-stage imputation method (EM) by 0%, 21.2%, 6.0%, 16.7%, 14.8%, 17.3%, 20.6%, and 20.2% in terms of ACC, and 0%, 22.2%, 5.0%, 19.0%, 15.1%, 21.6%, 26.3%, and 22.4% in terms of NMI, with the variation of missing ratios in $[0.1, 0.2, \ldots, 0.7]$ on the Seeds dataset. From Figures 3 and 4, we can observe that the trends of purity and F-score indicators are also the same. These results validate the effectiveness of unifying the imputation and clustering.

(2) Although the recently proposed algorithm, dynamically K-means filling clustering, achieves fairly good performance, Gaussian mixture model clustering is able to handle more complex multimodal data and achieves better performance. For example, the proposed algorithm improves the

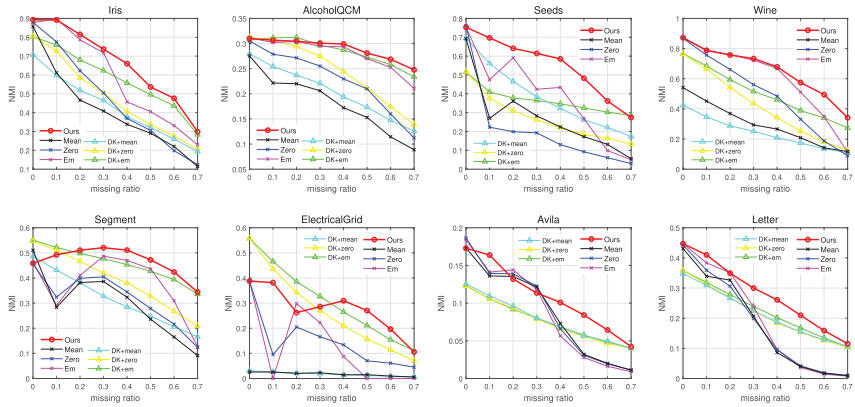

Fig. 2. NMI comparison with the variation of missing ratios on eight benchmark datasets.

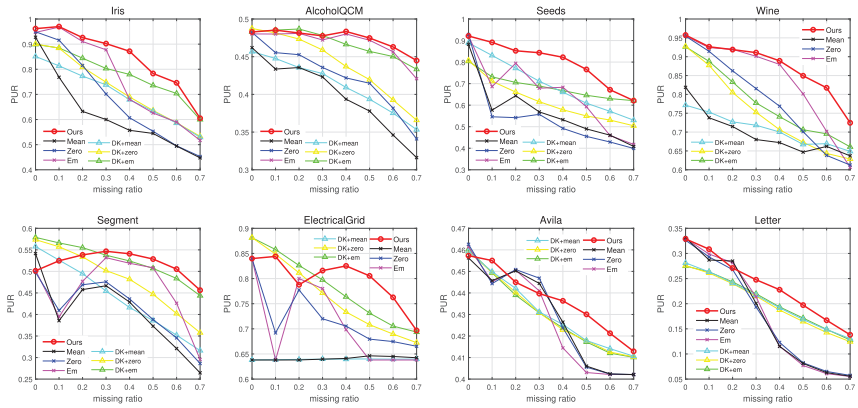

Fig. 3. PUR comparison with the variation of missing ratios on eight benchmark datasets.

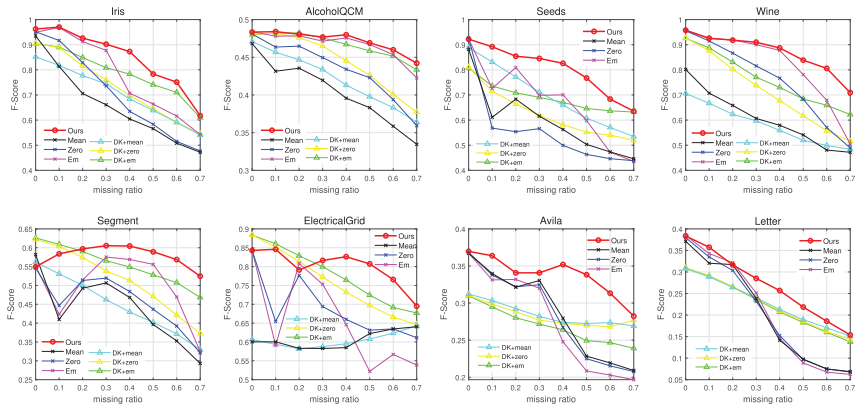

Fig. 4. F-score comparison with the variation of missing ratios on eight benchmark datasets.

Table 2. Aggregated ACC, NMI, F-score, and PUR Comparison (mean±std) of Different Algorithms on Eight Benchmark Datasets

Dataset	Mean	Zero	Em	DK+Mean	DK+Zero	DK+Em	Ours
ACC (%)							
Iris	61.3 ± 1.5	67.3 ± 1.5	76.0 ± 2.5	68.6 ± 1.0	70.7 ± 1.2	76.0 ± 0.8	**84.4 ± 2.2**
AlcoholQCM	38.0 ± 0.3	40.1 ± 0.5	46.1 ± 0.4	39.2 ± 0.3	41.8 ± 0.4	44.8 ± 0.4	**46.8 ± 0.4**
Seeds	56.6 ± 3.6	53.9 ± 1.3	64.7 ± 1.5	69.2 ± 0.8	61.3 ± 1.0	67.9 ± 0.6	**79.3 ± 1.0**
Wine	58.0 ± 1.6	74.8 ± 0.8	81.8 ± 0.6	56.8 ± 0.9	70.0 ± 0.6	75.2 ± 0.7	**87.0 ± 0.7**
Segment	39.2 ± 0.5	40.2 ± 0.5	44.7 ± 0.7	41.0 ± 0.2	45.5 ± 0.4	49.9 ± 0.3	**50.8 ± 0.3**
ElectricalGrid	58.8 ± 0.4	68.2 ± 2.0	64.8 ± 2.0	58.0 ± 0.5	75.2 ± 0.7	77.5 ± 0.4	**79.7 ± 0.6**
Avila	25.5 ± 0.2	25.1 ± 0.3	23.9 ± 0.3	24.3 ± 0.2	24.1 ± 0.2	23.0 ± 0.3	**29.1 ± 0.1**
Letter	16.5 ± 0.2	16.4 ± 0.1	16.7 ± 0.2	19.0 ± 0.1	18.8 ± 0.1	18.9 ± 0.1	**22.1 ± 0.1**
NMI (%)							
Iris	41.3 ± 1.2	47.2± 1.3	58.8 ± 2.7	43.0± 1.0	48.1 ± 1.6	58.0 ± 0.8	**66.3 ± 2.6**
AlcoholQCM	18.1 ± 0.4	22.8 ± 0.6	28.0 ± 0.5	20.4 ± 0.3	24.5 ± 0.5	28.5 ± 0.3	**29.0 ± 0.3**
Seeds	27.4 ± 4.9	21.1 ± 1.5	38.7 ± 1.7	38.6 ± 0.9	27.1 ± 1.2	36.6 ± 1.0	**55.1 ± 1.3**
wine	29.9 ± 1.8	49.3 ± 1.3	59.8 ± 1.0	24.2 ± 1.1	41.4 ± 1.0	50.3 ± 0.8	**65.5 ± 1.5**
Segment	29.7 ± 0.6	31.8 ± 0.6	37.4 ± 1.0	31.6 ± 0.2	39.1 ± 0.3	45.7 ± 0.3	**46.7 ± 0.5**
ElectricalGrid	1.76 ± 0.2	14.6 ± 2.3	12.5 ± 2.6	1.76 ± 0.2	27.0 ± 0.5	**31.0 ± 0.2**	27.5 ± 0.7
Avila	8.78 ± 0.1	8.93 ± 0.2	8.71 ± 0.2	7.82 ± 0.1	7.61 ± 0.1	7.69 ± 0.1	**10.92 ± 0.1**
Letter	18.2 ± 0.2	18.4 ± 0.2	19.7 ± 0.3	21.6 ± 0.1	21.6 ± 0.1	22.6 ± 0.1	**28.1± 0.1**
F-Score (%)							
Iris	65.8 ± 1.0	70.5 ± 1.1	78.0 ± 2.2	70.6 ± 0.7	73.1 ± 1.0	78.7 ± 0.5	**84.8 ± 2.0**
AlcoholQCM	40.3 ± 0.3	43.4 ± 0.5	46.6 ± 0.3	42.1 ± 0.3	44.4 ± 0.3	46.6 ± 0.3	**47.2 ± 0.4**
Seeds	59.7 ± 3.3	55.8 ± 1.1	67.0 ± 1.5	69.7 ± 0.7	62.5 ± 0.9	69.1 ± 0.7	**80.3 ± 0.7**
Wine	60.6 ± 1.2	75.8 ± 0.7	81.8 ± 0.7	58.2 ± 0.8	71.4 ± 0.6	76.4 ± 0.6	**86.9 ± 0.7**
Segment	43.8 ± 0.5	45.7 ± 0.5	49.8 ± 1.0	44.8 ± 0.2	51.5 ± 0.3	55.5 ± 0.3	**57.8 ± 0.4**
ElectricalGrid	60.6 ± 0.4	68.8 ± 1.6	65.9 ± 1.5	60.5 ± 0.3	75.8 ± 0.5	77.9 ± 0.3	**79.9 ± 0.5**
Avila	28.7 ± 0.1	28.3 ± 0.3	27.6 ± 0.3	28.5 ± 0.2	28.3 ± 0.2	26.9 ± 0.3	**33.8 ± 0.1**
Letter	20.4 ± 0.2	20.5 ± 0.1	20.8 ± 0.3	22.8 ± 0.1	22.6 ± 0.1	22.4 ± 0.1	**27.0 ± 0.1**
PUR (%)							
Iris	62.1 ± 1.5	68.6 ± 1.3	76.4 ± 2.4	70.0 ± 0.8	72.3 ± 1.1	78.1 ± 0.6	**84.6 ± 2.1**
AlcoholQCM	39.9 ± 0.3	42.3 ± 0.5	46.8 ± 0.4	41.2 ± 0.3	44.0 ± 0.4	46.8 ± 0.3	**47.4 ± 0.4**
Seeds	57.0 ± 3.6	54.4 ± 1.3	65.4 ± 1.4	69.7 ± 0.7	62.0 ± 1.0	68.7 ± 0.7	**79.9 ± 0.9**
Wine	69.6 ± 1.3	78.3 ± 0.5	83.6 ± 0.5	70.7 ± 0.5	75.2 ± 0.5	77.8 ± 0.6	**87.4 ± 0.7**
Segment	40.5 ± 0.4	41.3 ± 0.5	45.7 ± 0.7	43.8 ± 0.2	48.2 ± 0.3	**52.5 ± 0.2**	51.8 ± 0.3
ElectricalGrid	64.1 ± 0.1	71.9 ± 1.3	70.9 ± 1.4	63.9 ± 0.1	76.5 ± 0.5	78.2 ± 0.3	**79.7 ± 0.5**
Avila	42.9 ± 0.1	43.0 ± 0.1	42.7 ± 0.1	43.1 ± 0.1	43.0 ± 0.1	43.0 ± 0.1	**43.7 ± 0.0**
Letter	17.7 ± 0.2	17.6 ± 0.1	17.9 ± 0.2	20.5 ± 0.1	20.1 ± 0.1	20.5 ± 0.1	**23.6 ± 0.1**

second-best method (DK+mean) by 3.3%, 6.1%, 7.8%, 13.0%, 16.1%, 15.8%, 9.9%, and 9.3% in terms of ACC, and 5.8%, 13.6%, 17.6%, 23.0%, 26.3%, 22.0%, 14.1%, and 10.1% in terms of NMI, with the variation of missing ratios in $[0.1, 0.2, \ldots, 0.7]$ on the Seeds dataset. It can be observed from Figures 3 and 4 that the trends of purity and F-score indicators are also the same. These results verify the effectiveness of the Gaussian mixture model rather than the K-means method.

(3) When the missing ratio exceeds 40%, the effect of the existing two-stage imputation methods drops significantly. However, the proposed algorithm has the best robustness compared to other

algorithms, and can maintain the best performance under the condition of an increasing missing ratio.

(4) We also found a phenomenon worthy of consideration: some of the proposed algorithm's clustering results are better when the missing ratio is larger than when it is lower, even to zero. For example, the ACC value on the Segment dataset is about 4% higher when the missing rate reaches 30% than when it is zero, and the trends of other indicators are also the same. It seems that the algorithm performs better with incomplete data than complete data, which is not in line with intuitive logic. We repeated the experiment many times with the code being correct and verified the authenticity of the result. After research, we believe that there are many noises and some natural outliers based on the data itself, which may affect the noise points when dealing with missing data, so the clustering result of the algorithm may be better than the case of complete data. Based on this, in future research, we can consider how to use this point to deal with the noise problem of large-scale data.

We also report the aggregated performance metrics (ACC, NMI, F-score, and PUR) and the standard deviation in Table 2, where the best results are shown in bold. Again, we observe that the proposed algorithm almost always achieves the state of the art on each performance metric on all eight benchmark datasets. For example, the proposed algorithm exceeds the second-best one (DK+em) by 8.4%, 11.4%, 11.8%, and 6.1% in terms of ACC, and by 8.3%, 18.5%, 15.2%, and 3.2% in terms of NMI on the Iris, Seeds, Wine, and Avila datasets, respectively. Compared with the traditional Gaussian mixture model clustering algorithm, the proposed algorithm's clustering performance also has huge advantages. For example, the proposed algorithm exceeds the GMM with EM imputation (EM) by 8.4%, 14.6%, 5.2%, 6.1%, 14.9%, and 5.4% in terms of ACC; 6.8%, 13.3%, 5.1%, 8.0%, 14.0%, and 6.2% in terms of F-score; and by 8.2%, 14.5%, 3.8%, 6.1%, 8.8%, and 5.7% in terms of PUR on the Iris, Seeds, Wine, ElectricalGrid, and Avila datasets, respectively. These results are consistent with our observations in Figures 1, 2, 3, and 4 and validate the effectiveness of the proposed algorithm.

4.3 Advantages of Dynamically GMM Clustering

(1) The proposed algorithm adopts a Gaussian mixture model as the basic clustering model, which can address the complicated multi-modal data tasks better than others such as K-Means, Fuzzy CMeans, DBSCAN, and hierarchical clustering.

(2) The proposed algorithm unifies the imputation and clustering into a single optimization, and these two steps alternatively negotiate with each other to achieve optimum. By this way, the proposed algorithm makes full use of information from the missing entries and the imputed data can best serve for GMM clustering, resulting in better clustering performance.

4.4 Convergence

Our algorithm is outlined in Algorithm 1, where $obj^{(t)}$ denotes the objective value at the tth iteration. At each iteration, the objective value is monotonically increased when optimizing one variable with others fixed. At the same time, the EM algorithm is proven to converge when the likelihood function is monotonically increased. As a result, the proposed algorithm is theoretically guaranteed to converge to a local maximum. We also record the objective values with iterations and plot them in Figure 5. As observed, the objective value does monotonically increase at each iteration and it usually converges in fewer than 100 iterations.

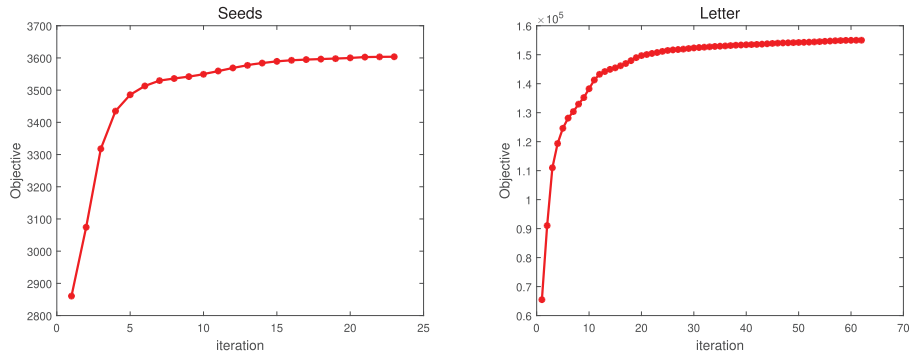

Fig. 5. The objective value with iterations performed on Seeds and Letter datasets.

5 CONCLUSION

While the recently proposed dynamically K-means is able to handle incomplete clustering, its inability to handle complicated multi-modal data prevents it from most practical clustering tasks. This article proposed a dynamically filling algorithm, GMM clustering with incomplete data, which integrates the imputation and GMM clustering into a unified learning procedure, to make full use of the information from the missing entries to serve clustering tasks better. The proposed alternative algorithm effectively and efficiently solves the resultant optimization problem, and the significantly improved clustering performance is demonstrated via extensive experiments on UCI benchmark datasets. In the future, we plan to apply the one-stage framework to other clustering tasks and extend our algorithm to a general framework and to one that can handle noise by dynamically clustering.

REFERENCES

[1] Elie Aljalbout, Vladimir Golkov, Yawar Siddiqui, and Daniel Cremers. 2018. Clustering with deep learning: Taxonomy and new methods. *arXiv: Learning* (2018).

[2] Marco Aste, Massimo Boninsegna, Antonino Freno, and Edmondo Trentin. 2015. Techniques for dealing with incomplete data: A tutorial and survey. *Pattern Analysis and Applications* 18, 1 (2015), 1–29.

[3] Steffen Bickel and Tobias Scheffer. 2004. Multi-view clustering. In *Fourth IEEE International Conference on Data Mining (ICDM'04)*. 19–26.

[4] Charles A. Bouman, Michael Shapiro, G. W. Cook, C. Brian Atkins, and Hui Cheng. 1997. *Cluster: An Unsupervised Algorithm for Modeling Gaussian Mixtures*. Technical Report.

[5] Magalie Celton, Alain Malpertuy, Gaëlle Lelandais, and Alexandre G. De Brevern. 2010. Comparative analysis of missing value imputation methods to improve clustering and interpretation of microarray experiments. *BMC Genomics* 11, 1 (2010), 15.

[6] Liang Du, Peng Zhou, Lei Shi, Hanmo Wang, Mingyu Fan, Wenjian Wang, and Yi-Dong Shen. 2015. Robust multiple kernel k-means clustering using ℓ_{21}-norm. In *International Joint Conference on Artificial Intelligence (IJCAI'15)*. 3476–3482.

[7] Eduard Eiben, Robert Ganian, Iyad Kanj, Sebastian Ordyniak, and Stefan Szeider. 2019. On clustering incomplete data. *arXiv preprint arXiv:1911.01465* (2019).

[8] Payam Ezatpoor, Justin Zhan, Jimmy Ming-Tai Wu, and Carter Chiu. 2018. Finding top-k dominance on incomplete big data using Mapreduce framework. *IEEE Access* 6 (2018), 7872–7887.

[9] Pedro J. García-Laencina, José-Luis Sancho-Gómez, Aníbal R. Figueiras-Vidal, and Michel Verleysen. 2009. K nearest neighbours with mutual information for simultaneous classification and missing data imputation. *Neurocomputing* 72, 7–9 (2009), 1483–1493.

[10] Zoubin Ghahramani and Michael I. Jordan. 1994. Supervised learning from incomplete data via an EM approach. In *Advances in Neural Information Processing Systems*. 120–127.

[11] Iffat A. Gheyas and Leslie S. Smith. 2010. A neural network-based framework for the reconstruction of incomplete data sets. *Neurocomputing* 73, 16–18 (2010), 3039–3065.

[12] Xifeng Guo, Long Gao, Xinwang Liu, and Jianping Yin. 2017. Improved deep embedded clustering with local structure preservation. In *International Joint Conference on Artificial Intelligence (IJCAI'17)*. 1753–1759.
[13] John A. Hartigan. 1975. *Clustering Algorithms*. John Wiley & Sons, Inc.
[14] Shao-Yuan Li, Yuan Jiang, and Zhi-Hua Zhou. 2014. Partial multi-view clustering. In *Association for the Advance of Artificial Intelligence (AAAI'14)*. 1968–1974.
[15] Tianhao Li, Liyong Zhang, Wei Lu, Hui Hou, Xiaodong Liu, Witold Pedrycz, and Chongquan Zhong. 2017. Interval kernel fuzzy C-means clustering of incomplete data. *Neurocomputing* 237 (2017), 316–331.
[16] Xinwang Liu, Yong Dou, Jianping Yin, Lei Wang, and En Zhu. 2016. Multiple kernel k-means clustering with matrix-induced regularization. In *Association for the Advance of Artificial Intelligence (AAAI'16)*. 1888–1894.
[17] Xinwang Liu, Wen Gao, Xinzhong Zhu, Miaomiao Li, Lei Wang, En Zhu, Tongliang Liu, Marius Kloft, Dinggang Shen, and Jianping Yin. 2020. Multiple kernel k-means with incomplete kernels. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42, 5 (2020), 1191–1204.
[18] Xinwang Liu, Lei Wang, Xinzhong Zhu, Miaomiao Li, En Zhu, Tongliang Liu, Li Liu, Yong Dou, and Jianping Yin. 2020. Absent multiple kernel learning algorithms. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42, 6 (2020), 1303–1316.
[19] Volodymyr Melnykov and Ranjan Maitra. 2010. Finite mixture models and model-based clustering. *Statistics Surveys* 4 (2010), 80–116.
[20] Erxue Min, Xifeng Guo, Qiang Liu, Gen Zhang, Jianjing Cui, and Jun Long. 2018. A survey of clustering with deep learning: From the perspective of network architecture. *IEEE Access* 6 (2018), 39501–39514.
[21] Chunfeng Song, Feng Liu, Yongzhen Huang, Liang Wang, and Tieniu Tan. 2013. Auto-encoder based data clustering. In *Iberoamerican Congress on Pattern Recognition (CIARP'13)*. 117–124.
[22] Anusua Trivedi, Piyush Rai, Hal Daumé III, and Scott L. DuVall. 2010. Multiview clustering with incomplete views. In *NIPS Workshop*, Vol. 224.
[23] Siwei Wang, Miaomiao Li, Ning Hu, En Zhu, Jingtao Hu, Xinwang Liu, and Jianping Yin. 2019. K-means clustering with incomplete data. *IEEE Access* 7 (2019), 69162–69171.
[24] Yang Wang, Xuemin Lin, Lin Wu, Wenjie Zhang, Qing Zhang, and Xiaodi Huang. 2015. Robust subspace clustering for multi-view data by exploiting correlation consensus. *IEEE Transactions on Image Processing* 24, 11 (2015), 3939–3949.
[25] Y. Wang, L. Wu, X. Lin, and J. Gao. 2018. Multiview spectral clustering via structured low-rank matrix factorization. *IEEE Transactions on Neural Networks and Learning Systems* 29, 10 (2018), 4833–4843.
[26] Lin Wu, Yang Wang, and Ling Shao. 2019. Cycle-consistent deep generative hashing for cross-modal retrieval. *IEEE Transactions on Image Processing* 28, 4 (2019), 1602–1612.
[27] Cong-Hua Xie, Jin-Yi Chang, and Yong-Jun Liu. 2013. Estimating the number of components in Gaussian mixture models adaptively for medical image. *Optik* 124, 23 (2013), 6216–6221.
[28] Junyuan Xie, Ross Girshick, and Ali Farhadi. 2016. Unsupervised deep embedding for clustering analysis. In *International Conference on Machine Learning (ICML'16)*. 478–487.
[29] Rui Xu and Donald Wunsch. 2005. Survey of clustering algorithms. *IEEE Transactions on Neural Networks* 16, 3 (2005), 645–678.
[30] Shi Yu, Léon-Charles Tranchevent, Xinhai Liu, Wolfgang Glänzel, Johan A. K. Suykens, Bart De Moor, and Yves Moreau. 2012. Optimized data fusion for kernel k-means clustering. *IEEE TPAMI* 34, 5 (2012), 1031–1039.
[31] En Zhu, Sihang Zhou, Yueqing Wang, Jianping Yin, Miaomiao Li, Xinwang Liu, and Yong Dou. 2017. Optimal neighborhood kernel clustering with multiple kernels. In *Association for the Advance of Artificial Intelligence (AAAI'17)*. 2266–2272.

Received April 2020; revised May 2020; accepted June 2020